

P–SV–SH Optical Theorem: Forward Amplitude, Flux Weights and Extinction

T. M. Nestor

Research School of Earth Sciences, Australian National University

Status. Explanatory note. The results of §§1–3 are textbook material (Varatharajulu, 1977; Pao and Mow, 1973; Newton, 1982). In this project the S-incidence form (13) has been checked numerically for the lossless sphere to 10^{-8} (FirstOrderContrastOperator.tex, toroidal-term paragraph). The P-incidence form (12) is checked for the exact Mie sphere and for the T_{27} cube (§8). The cube symmetry statements of §4 and the T-matrix form of §5 have no independent check here and are *unvalidated* in the sense of the project’s engineering standards.

1 Setting and far field

A bounded scatterer V is embedded in a homogeneous isotropic elastic background with density ρ and wave speeds α (P) and β (S). The time dependence is $e^{-i\omega t}$ throughout, and

$$k_P = \omega/\alpha, \quad k_S = \omega/\beta, \quad k_P < k_S. \quad (1)$$

Outside V the scattered displacement satisfies the Kupradze radiation condition (Pao and Varatharajulu, 1976). Far from V it separates into an outgoing longitudinal wave and an outgoing transverse wave (Pao and Mow, 1973; Aki and Richards, 2002):

$$\mathbf{u}^s(\mathbf{x}) \sim \mathbf{A}_P(\hat{\mathbf{x}}) \frac{e^{ik_P r}}{r} + \mathbf{A}_S(\hat{\mathbf{x}}) \frac{e^{ik_S r}}{r}, \quad r = |\mathbf{x}| \rightarrow \infty, \quad (2)$$

where $\mathbf{A}_P \parallel \hat{\mathbf{x}}$ and $\mathbf{A}_S \perp \hat{\mathbf{x}}$. The transverse amplitude splits in the spherical triad into

$$\mathbf{A}_S = A_{SV} \hat{\boldsymbol{\theta}} + A_{SH} \hat{\boldsymbol{\phi}}, \quad (3)$$

so the far field carries three channels, P, SV and SH. The amplitudes $\mathbf{A}_P$ and $\mathbf{A}_S$ are displacement amplitudes with dimension of length, per unit incident amplitude.

Two incident waves are considered, each of unit displacement amplitude:

$$\text{P incidence: } \mathbf{u}^i = \hat{\mathbf{d}} e^{ik_P \hat{\mathbf{d}} \cdot \mathbf{x}}, \quad (4)$$

$$\text{S incidence: } \mathbf{u}^i = \hat{\mathbf{p}} e^{ik_S \hat{\mathbf{d}} \cdot \mathbf{x}}, \quad \hat{\mathbf{p}} \cdot \hat{\mathbf{d}} = 0, \quad |\hat{\mathbf{p}}| = 1. \quad (5)$$

For S incidence $\hat{\mathbf{p}}$ may be the SV or SH direction, or complex for elliptical polarisation.

2 Energy flux and channel weights

For time-harmonic motion the time-averaged energy flux vector is

$$\langle \mathbf{J} \rangle = \frac{\omega}{2} \text{Im}(\boldsymbol{\sigma} \cdot \bar{\mathbf{u}}), \quad (6)$$

with $\boldsymbol{\sigma}$ the stress tensor (Achenbach, 1973; Aki and Richards, 2002). For a plane wave of amplitude U travelling at speed c this reduces to $\frac{1}{2}\rho\omega^2 c |U|^2$ along the propagation direction. Applied to each term of (2), the scattered power per unit solid angle is

$$\frac{dP_s}{d\Omega} = \frac{1}{2}\rho\omega^2 (\alpha |\mathbf{A}_P|^2 + \beta |\mathbf{A}_S|^2). \quad (7)$$

The P-S cross term drops out of (7): its phase $e^{i(k_S - k_P)r}$ oscillates in r , and its polarisations are orthogonal.

Cross sections are powers divided by the incident flux, which is $\frac{1}{2}\rho\omega^2\alpha$ for (4) and $\frac{1}{2}\rho\omega^2\beta$ for (5). Each scattered channel is therefore weighted by the ratio of its wave speed to the incident speed (Wu and Aki, 1985; Sato et al., 2012):

$$\text{P incidence: } \sigma_{\text{sc}} = \oint |\mathbf{A}_P|^2 d\Omega + \frac{\beta}{\alpha} \oint |\mathbf{A}_S|^2 d\Omega, \quad (8)$$

$$\text{S incidence: } \sigma_{\text{sc}} = \frac{\alpha}{\beta} \oint |\mathbf{A}_P|^2 d\Omega + \oint |\mathbf{A}_S|^2 d\Omega. \quad (9)$$

Since $\beta/\alpha = k_P/k_S$, the weights are ratios of wavenumbers too. The S term in (8) and (9) is the sum of its SV and SH parts, $|\mathbf{A}_S|^2 = |A_{SV}|^2 + |A_{SH}|^2$.

3 The theorem

Write the total field as $\mathbf{u} = \mathbf{u}^i + \mathbf{u}^s$ and integrate the flux (6) over a large sphere S_R . The net outward flux is $-P_{\text{abs}}$, where P_{abs} is the power absorbed in V . The incident term alone integrates to zero. The scattered term gives P_s through (7). What remains is the interference term,

$$P_{\text{ext}} = -\frac{\omega}{2} \oint_{S_R} \text{Im}(\boldsymbol{\sigma}^i \cdot \bar{\mathbf{u}}^s + \boldsymbol{\sigma}^s \cdot \bar{\mathbf{u}}^i) \cdot \hat{\mathbf{x}} dS, \quad (10)$$

with the balance

$$P_{\text{ext}} = P_s + P_{\text{abs}}, \quad \sigma_{\text{ext}} = \sigma_{\text{sc}} + \sigma_{\text{abs}}. \quad (11)$$

The integrand of (10) carries the phase $e^{ik(r - \hat{\mathbf{d}} \cdot \mathbf{x})}$, which oscillates rapidly on S_R as $R \rightarrow \infty$. Stationary phase (Morse and Feshbach, 1953) confines the integral to the forward point $\hat{\mathbf{x}} = \hat{\mathbf{d}}$, where the phase is stationary and independent of R . The backward point contributes a term proportional to e^{2ikR} . Equation (11) must hold for every R , so that term vanishes. This is the same argument as in the scalar and electromagnetic cases (Newton, 1976; Bohren and Huffman, 1983).

The elastic case adds one step. The incident wave can interfere only with the scattered wave of the *same* type:

- **Different wavenumbers.** Incident P interferes with scattered S with the phase $e^{i(k_S - k_P)R}$ at the forward point, and the reverse pairing is the same. This phase is never stationary in R , so the term vanishes by the argument above.
- **Orthogonal polarisations.** At $\hat{\mathbf{x}} = \hat{\mathbf{d}}$ the scattered S wave is transverse, $\mathbf{A}_S(\hat{\mathbf{d}}) \perp \hat{\mathbf{d}}$, while an incident P wave is polarised along $\hat{\mathbf{d}}$. Conversely, a scattered P wave is polarised along $\hat{\mathbf{d}}$, while an incident S wave is polarised along $\hat{\mathbf{p}} \perp \hat{\mathbf{d}}$. The leading-order cross flux is proportional to the polarisation product and is zero.

Evaluating the same-mode forward term gives the elastic optical theorem, or forward-amplitude theorem (Varatharajulu, 1977):

$$\text{P incidence: } \sigma_{\text{ext}} = \frac{4\pi}{k_P} \text{Im}[\hat{\mathbf{d}} \cdot \mathbf{A}_P(\hat{\mathbf{d}})], \quad (12)$$

$$\text{S incidence: } \sigma_{\text{ext}} = \frac{4\pi}{k_S} \text{Im}[\hat{\mathbf{p}} \cdot \mathbf{A}_S(\hat{\mathbf{d}})]. \quad (13)$$

For a lossless scatterer $\sigma_{\text{abs}} = 0$, so the imaginary part of one forward amplitude sets the total power scattered into all three channels P, SV and SH, with the weights of (8) and (9). Mode conversion is fully accounted for, but only on the right of (11). It never appears in σ_{ext} .

Table 1. Contents of the two sides of (11) for each incident type. “Forward” means $\hat{\mathbf{x}} = \hat{\mathbf{d}}$.

Incident	σ_{ext} uses	σ_{sc} uses
P	forward P amplitude, component along $\hat{\mathbf{d}}$	$ \mathbf{A}_P ^2$ with weight 1; $ A_{SV} ^2 + A_{SH} ^2$ with weight β/α
SV	forward S amplitude, $\hat{\theta}$ component	$ \mathbf{A}_P ^2$ with weight α/β ; $ A_{SV} ^2 + A_{SH} ^2$ with weight 1
SH	forward S amplitude, $\hat{\phi}$ component	as for SV

4 The forward S matrix and polarisation

For S incidence the forward response is linear in $\hat{\mathbf{p}}$. In the $(\hat{\theta}, \hat{\phi})$ basis at $\hat{\mathbf{x}} = \hat{\mathbf{d}}$ it is a 2×2 matrix $\mathbf{F}(\hat{\mathbf{d}})$,

$$\mathbf{A}_S(\hat{\mathbf{d}}) = \mathbf{F}(\hat{\mathbf{d}}) \hat{\mathbf{p}}, \quad \sigma_{\text{ext}}(\hat{\mathbf{p}}) = \frac{4\pi}{k_S} \text{Im}[\hat{\mathbf{p}}^\dagger \mathbf{F}(\hat{\mathbf{d}}) \hat{\mathbf{p}}]. \quad (14)$$

Off-diagonal elements of $\mathbf{F}$ (forward SV→SH conversion) enter σ_{ext} only through the anti-Hermitian part of $\mathbf{F}$. For linear SV or SH polarisation, the extinction takes the corresponding diagonal element. Forward cross-polarised power still counts in σ_{sc} through $|\mathbf{A}_S|^2$.

The structure of $\mathbf{F}$ follows from the point symmetry of the scatterer about the axis $\hat{\mathbf{d}}$. It must commute with every symmetry operation that fixes $\hat{\mathbf{d}}$.

- **Sphere.** Full axial symmetry, so $\mathbf{F}$ is scalar, and SV and SH extinction are equal for any $\hat{\mathbf{d}}$.
- **Cube, $\hat{\mathbf{d}}$ along [100] or [111].** The symmetry groups are C_{4v} and C_{3v} . Each contains a rotation of order at least three and a mirror containing the axis, and this forces $\mathbf{F}$ to be scalar. Extinction is independent of polarisation.
- **Cube, $\hat{\mathbf{d}}$ along [110].** The group is C_{2v} , with mirrors (001) and (110). In the basis aligned with them, $\mathbf{F}$ is diagonal with two different eigenvalues in general. The two linear polarisations have different extinction, so a cube is weakly dichroic, and weakly birefringent, along face diagonals.
- **Cube, general $\hat{\mathbf{d}}$.** No symmetry. $\mathbf{F}$ is a general 2×2 matrix, and the eigenpolarisations need not be linear.

The same argument applied to P incidence gives nothing new, since $\hat{\mathbf{d}} \cdot \mathbf{A}_P(\hat{\mathbf{d}})$ is a scalar for every scatterer.

5 T-matrix form: flux-normalised unitarity

In a spherical-wave basis the regular and outgoing vector wavefunctions split into a longitudinal family $\mathbf{L}$ (P), and two transverse families, $\mathbf{M}$ (toroidal, SH-type) and $\mathbf{N}$ (spheroidal, SV-type) (Pao and Mow, 1973; Waterman, 1976). The T-matrix maps regular incident coefficients to outgoing scattered coefficients. The scattering matrix

$$\mathbf{S} = \mathbf{I} + 2\mathbf{T} \quad (15)$$

maps incoming to outgoing waves (Varatharajulu and Pao, 1976; Newton, 1982).

If each basis function carries unit energy flux, the energy balance (11) becomes, for all incident vectors,

$$\mathbf{I} - \mathbf{S}^\dagger \mathbf{S} \geq 0, \quad \text{with equality when the scatterer is lossless.} \quad (16)$$

Substituting (15) gives, in the lossless case,

$$\mathbf{T} + \mathbf{T}^\dagger = -2 \mathbf{T}^\dagger \mathbf{T}, \quad -\text{Re} T_{\nu\nu} = \sum_{\mu} |T_{\mu\nu}|^2, \quad (17)$$

where μ runs over every outgoing channel in the $\mathbf{L}$, $\mathbf{M}$ and $\mathbf{N}$ families. Equation (17) is the optical theorem written one partial wave at a time: the left side is extinction, the right side is scattering. Summing over the partial waves of a plane wave and evaluating the result in the forward direction recovers (12) and (13) (Waterman, 1976; Varatharajulu, 1977).

Normalisation is essential. Standard vector wavefunctions are not flux-normalised. The $\mathbf{L}$ family is built on k_P and the $\mathbf{M}$, $\mathbf{N}$ families on k_S , so a raw T-matrix satisfies (17) only with channel weights. These weights are the same speed ratios as in (8) and (9). Without them, the P-S blocks appear to violate energy conservation by factors of β/α or α/β . The same rescaling

makes the reciprocity relation a plain transpose, $\mathbf{S} = \mathbf{S}^\top$ (Varatharajulu, 1977; Pao, 1978). The analogous statement for plane-wave interface coefficients is in `psvsh_unitarity.tex`.

6 Coherent attenuation

The theorem is what connects a single-scatterer T-matrix to the attenuation of the coherent wave in a random distribution. For number density n in the Foldy approximation (Foldy, 1945; Lax, 1951; Martin, 2006), the coherent P wave has effective wavenumber

$$K_p^2 \approx k_p^2 + 4\pi n \hat{\mathbf{d}} \cdot \mathbf{A}_p(\hat{\mathbf{d}}), \quad (18)$$

so to first order in n ,

$$\text{Im } K_p \approx \frac{2\pi n}{k_p} \text{Im}[\hat{\mathbf{d}} \cdot \mathbf{A}_p(\hat{\mathbf{d}})] = \frac{1}{2} n \sigma_{\text{ext}}. \quad (19)$$

The coherent amplitude therefore decays at half the extinction per unit length, and the coherent intensity decays at the full rate $n\sigma_{\text{ext}}$ (Sato et al., 2012). The S case is identical, with $\mathbf{F}(\hat{\mathbf{d}})$ taking the place of the scalar forward amplitude. Along a direction where $\mathbf{F}$ is not scalar, such as [110] for aligned cubes, the coherent S wave splits into two eigenpolarisations with different K_s .

7 Uses as a numerical check

For a lossless scatterer, the ratio $\sigma_{\text{ext}}/\sigma_{\text{sc}}$ is unity with no free parameters. The two sides are computed from different data: σ_{ext} from one forward amplitude, σ_{sc} from an angular quadrature over all three channels. The ratio therefore checks any T-matrix independently of how the T-matrix was obtained. Its main uses are these.

- **Truncation.** A deficit that shrinks with increasing n_{max} measures multipole truncation.
- **Mode-conversion weights.** Dropping the flux weight in (8) or (9) gives an error that grows with the converted power. It is a common failure mode.
- **Signs.** The theorem fixes the sign of a scattered amplitude but not its magnitude. For example it fixes the sign of the toroidal coefficient in `FirstOrderContrastOperator.tex`. A family that conserves energy on its own passes the test whether it is present or not.
- **Approximate T-matrices.** A Born or first-order T-matrix is real to leading order and has no radiation damping. Its forward amplitude has an imaginary part only at second order in contrast, so it cannot satisfy the theorem exactly (Gubernatis et al., 1977). The residual measures the missing self-consistency and is not a coding error.

8 Conventions and checks in this repository

Equations (12) and (13) assume $e^{-i\omega t}$ and far-field amplitudes defined by (2). Under $e^{+i\omega t}$ the imaginary part changes sign, and so does any convention that puts an extra minus sign on

the far-field amplitude. Every checker in `scattered_field.py` now uses the standard sign, $\sigma_{\text{ext}} = +(4\pi/k) \text{Im } f(\hat{\mathbf{d}})$, as does `docs/cube_tmatrix_closedform.tex`. An earlier minus sign there compensated a force-term sign error in the cube far field, and the two were removed together.

For P incidence on the lossless T_{27} cube, $\sigma_{\text{ext}}/\sigma_{\text{sc}}$ is 0.99997 at $ka = 0.05$ and 0.9986 at $ka = 0.3$. For a pure-modulus contrast it is 1.00002. Getting there required the modulus radiation reaction to be treated like the density one:

- **Kept in the far field.** At forward incidence the incident strain $ik_p \hat{\mathbf{d}} \hat{\mathbf{d}}$ is imaginary, so $\text{Im } \Delta \mathbf{c}^*$ is exactly what enters $\text{Im } f_p(\hat{\mathbf{d}})$. A far field built from $\text{Re } \Delta \mathbf{c}^*$ alone gave a pure-modulus cube $\sigma_{\text{ext}} = 0$ and the full cube a ratio of 0.61.
- **Built from the effective contrast.** The reaction is the radiated power of the dipole the far field actually emits, $\text{Im } \Delta \mathbf{c}^* = -(2V/\omega) \text{Re } \Delta \mathbf{c}^* : \mathbf{K} : \text{Re } \Delta \mathbf{c}^*$, with $\mathbf{K}$ the isotropic radiated-power kernel. Built from the bare contrast it overshoots, giving a ratio near 1.02. The exact Mie sphere obeys this identity with its own effective contrasts to 10^{-4} .

The closed form and its checks are in `docs/cube_tmatrix_closedform.tex` (eq. modulus-reaction). This case belongs with the *signs* and *mode-conversion* entries of §7: a missing radiation reaction is a bookkeeping defect, and the theorem detects it exactly.

References

- J. D. Achenbach. *Wave Propagation in Elastic Solids*, volume 16 of *North-Holland Series in Applied Mathematics and Mechanics*. North-Holland, Amsterdam, 1973.
- K. Aki and P. G. Richards. *Quantitative Seismology*. University Science Books, Sausalito, California, 2nd edition, 2002.
- C. F. Bohren and D. R. Huffman. *Absorption and Scattering of Light by Small Particles*. Wiley, New York, 1983.
- L. L. Foldy. The multiple scattering of waves. I. General theory of isotropic scattering by randomly distributed scatterers. *Physical Review*, 67:107–119, 1945. doi: 10.1103/PhysRev.67.107.
- J. E. Gubernatis, E. Domany, and J. A. Krumhansl. Formal aspects of the theory of the scattering of ultrasound by flaws in elastic materials. *Journal of Applied Physics*, 48:2804–2811, 1977. doi: 10.1063/1.324141.
- M. Lax. Multiple scattering of waves. *Reviews of Modern Physics*, 23:287–310, 1951. doi: 10.1103/RevModPhys.23.287.
- P. A. Martin. *Multiple Scattering: Interaction of Time-Harmonic Waves with N Obstacles*, volume 107 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 2006.

- P. M. Morse and H. Feshbach. *Methods of Theoretical Physics*, volume II. McGraw-Hill, New York, 1953.
- R. G. Newton. Optical theorem and beyond. *American Journal of Physics*, 44(7):639–642, 1976. doi: 10.1119/1.10324.
- R. G. Newton. *Scattering Theory of Waves and Particles*. Springer-Verlag, New York, 2nd edition, 1982.
- Y.-H. Pao. Betti’s identity and transition matrix for elastic waves. *Journal of the Acoustical Society of America*, 64(1):302–310, 1978. doi: 10.1121/1.381986.
- Y.-H. Pao and C. C. Mow. *Diffraction of Elastic Waves and Dynamic Stress Concentrations*. Crane Russak, New York, 1973.
- Y.-H. Pao and V. Varatharajulu. Huygens’ principle, radiation conditions, and integral formulas for the scattering of elastic waves. *Journal of the Acoustical Society of America*, 59(6):1361–1371, 1976.
- H. Sato, M. C. Fehler, and T. Maeda. *Seismic Wave Propagation and Scattering in the Heterogeneous Earth*. Springer, Berlin, 2nd edition, 2012. doi: 10.1007/978-3-642-23029-5.
- V. Varatharajulu. Reciprocity relations and forward amplitude theorems for elastic waves. *Journal of Mathematical Physics*, 18(4):537–543, 1977.
- V. Varatharajulu and Y.-H. Pao. Scattering matrix for elastic waves. I. Theory. *Journal of the Acoustical Society of America*, 60(3):556–566, 1976. doi: 10.1121/1.381129.
- P. C. Waterman. Matrix theory of elastic wave scattering. *Journal of the Acoustical Society of America*, 60(3):567–580, 1976.
- R.-S. Wu and K. Aki. Scattering characteristics of elastic waves by an elastic heterogeneity. *Geophysics*, 50(4):582–595, 1985. doi: 10.1190/1.1441934.